Timeline:
1 Nov: Leaderboard opens
11 Dec: Phase 1 final deadline
4 Jan: Phase 2 starts
28 Feb: Phase 2 final deadline

Dataset:

1. **ASMS revealed labels:** 39 positives + ~6,300 negatives
   CatalogID Label RDKit_SMILES
   Z905549638 1 O=C(Nc1cccc(-n2ccnc2)c1)c1ccc(-c2ccco2)[nH]c1=O
2. **ASMS no labels:** ~185k rows
   CatalogID SMILES
   Z1840178734 CCc1c(C(=O)N(C)CC2CCOC2)cnn1-c1ccccc1Cl

Validation/Testing:
Chemical-similarity compounds grouping (closely related chemical families)
K-fold cross-validation (3 to 4?)

Workflow:
Compound structure -> chemical features -> selection of descriptors ->
Quantum circuit -> measured features -> classifier -> hit score

Chemical features (RDKit https://www.rdkit.org/docs/GettingStartedInPython.html)
Molecular fingerprints, weight, hydrogen-bond counts, and predicted lipophilicity

A classical reference as benchmark: random forest / logistic regression
0.83-0.86 Accuracy
Quantum candidates:

A. Projected quantum features: https://www.nature.com/articles/s41467-021-22539-9
   Encode compressed molecular descriptors
B. Landmark quantum kernel: https://link.springer.com/article/10.1007/s42484-023-00131-2
   Estimate molecular similarities (Nystrom approximation)
C. Quantum Approximate Optimization Algorithm: https://arxiv.org/abs/2211.02861
   Quantum optimization select the molecular descriptors
D. Hybrid models